

Scenario	Validation method	Balanced Accuracy	Macro F1	MCC	ROC AUC	Per-class Recall (Low/High)
Baseline (within)	Stratified K-fold(Full A)	0.67	0.67	0.35	0.74	0.64/0.70
	Stratified K-fold(Full B)	0.68	0.65	0.36	0.75	0.60/0.76
	Stratified K-fold(B: Stroop)	0.69	0.68	0.40	0.76	0.68/0.70
	Stratified K-fold(B: Arithmetic)	0.70	0.68	0.41	0.76	0.69/0.70
	Stratified K-fold(B: Mirror)	0.70	0.70	0.43	0.77	0.72/0.69
Cross-subject	Leave-one-subject-out(Full A)	0.66	0.65	0.33	0.74	0.65/0.67
	Leave-one-subject-out(Full B)	0.65	0.61	0.31	0.73	0.58/0.72
Cross-task (Dataset B)	Full B $\rightarrow$ Stroop	0.64	0.61	0.31	0.71	0.60/0.68
	Full B $\rightarrow$ Arithmetic	0.65	0.63	0.33	0.71	0.59/0.71
	Full B $\rightarrow$ Mirror	0.67	0.65	0.37	0.77	0.58/0.77
	Stroop $\rightarrow$ Full B	0.61	0.54	0.22	0.67	0.63/0.58
	Stroop $\rightarrow$ Arithmetic	0.56	0.52	0.13	0.58	0.67/0.45
	Stroop $\rightarrow$ Mirror	0.62	0.59	0.27	0.69	0.65/0.59
	Arithmetic $\rightarrow$ Full B	0.64	0.56	0.25	0.70	0.68/0.59
	Arithmetic $\rightarrow$ Stroop	0.58	0.55	0.18	0.62	0.68/0.48
	Arithmetic $\rightarrow$ Mirror	0.67	0.64	0.36	0.74	0.66/0.67
	Mirror $\rightarrow$ Full B	0.65	0.58	0.28	0.73	0.71/0.59
	Mirror $\rightarrow$ Stroop	0.61	0.58	0.25	0.67	0.72/0.51
	Mirror $\rightarrow$ Arithmetic	0.64	0.62	0.32	0.72	0.70/0.59
Cross-dataset	Full A $\rightarrow$ Full B	0.52	0.49	0.04	0.52	0.28/0.75
	Full B $\rightarrow$ Full A	0.54	0.52	0.09	0.55	0.37/0.71
	Full A $\rightarrow$ Stroop	0.48	0.42	-0.04	0.47	0.26/0.70
	Stroop $\rightarrow$ Full A	0.51	0.48	0.02	0.50	0.32/0.69
	Full A $\rightarrow$ Arithmetic	0.54	0.47	0.10	0.55	0.27/0.81
	Arithmetic $\rightarrow$ Full A	0.59	0.58	0.18	0.62	0.63/0.55
	Full A $\rightarrow$ Mirror	0.51	0.45	0.05	0.52	0.27/0.76
	Mirror $\rightarrow$ Full A	0.53	0.52	0.07	0.56	0.61/0.45